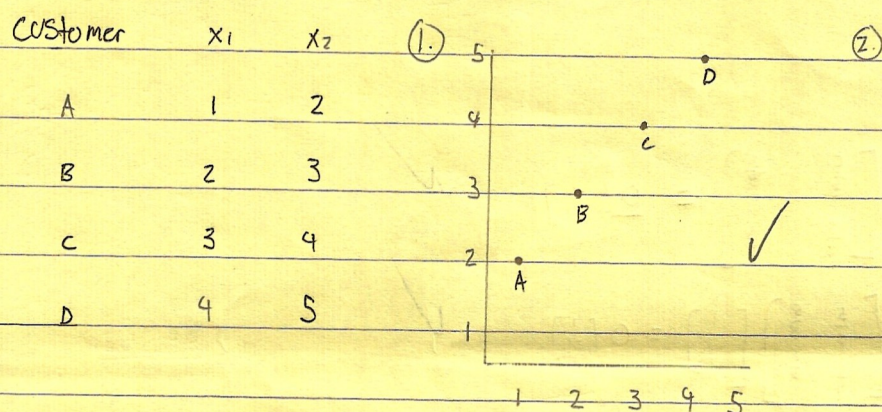




$$x_1 = \frac{1+2+3+4}{4} = \frac{10}{4} = 2.5 \quad \checkmark$$

$$x_2 = \frac{2+3+4+5}{4} = \frac{14}{4} = 3.5 \quad \checkmark$$

$$\textcircled{3} \begin{bmatrix} 1-2.5 & 2-3.5 \\ 2-2.5 & 3-3.5 \\ 3-2.5 & 4-3.5 \\ 4-2.5 & 5-3.5 \end{bmatrix}$$

$$\textcircled{4} \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$

= Acentered

$$A^T_{\text{centered}} = \begin{bmatrix} -1.5 & -0.5 & 0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \end{bmatrix} \quad \checkmark$$

$$\textcircled{5} A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \quad \frac{1}{4-1} A^T A = \begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} \quad \checkmark$$

$$\textcircled{6} \det(A - \lambda I) = 0$$

$$\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\det \left(\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$$

$$\det |A| = (a \times d) - (b \times c)$$

$$x_1 = \frac{10}{3} \quad x_2 = 0 \quad \checkmark \quad \checkmark$$

$$\begin{vmatrix} \frac{5}{3} - \lambda & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} - \lambda \end{vmatrix} = 0$$

$$= \left(\frac{5}{3} - \lambda \right)^2 - \left(\frac{5}{3} \right)^2$$

$$\lambda^2 + \frac{10}{3} \lambda = 0$$

$$\left(\frac{5}{3} - \lambda \right) \left(\frac{5}{3} - \lambda \right) - \left(\frac{5}{3} \right)^2$$

$$(\lambda + 0) \left(\lambda + \frac{10}{3} \right)$$

$$\left(\frac{5}{3} \right)^2 + 2 \left(\frac{5}{3} \lambda \right) + \lambda^2 - \left(\frac{5}{3} \right)^2$$

$$\frac{25}{9} + \frac{10}{3} \lambda + \lambda^2 - \frac{25}{9} = 0$$

$$\textcircled{7} (A - \lambda I) v = 0$$

$$\left(\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} - \begin{bmatrix} \frac{10}{3} & 0 \\ 0 & \frac{10}{3} \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \quad \begin{bmatrix} -\frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & -\frac{5}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -\frac{5}{3} v_1 + \frac{5}{3} v_2 \\ \frac{5}{3} v_1 - \frac{5}{3} v_2 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} v_2 = \frac{5}{3} v_1 \\ \frac{5}{3} v_1 = \frac{5}{3} v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \checkmark$$

$$\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \quad \begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} v_1 + \frac{5}{3} v_2 \\ \frac{5}{3} v_1 + \frac{5}{3} v_2 \end{bmatrix} = 0 = \begin{bmatrix} \frac{5}{3} v_2 = \frac{5}{3} v_1 \\ \frac{5}{3} v_1 = \frac{5}{3} v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \checkmark$$

8) $\lambda_1 = \frac{10}{3}$ $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\|v\| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$

$v_{\text{normal}} = \frac{v}{\|v\|} = \frac{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ✓

9) $Av = \lambda v$ $\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{10}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} = 0$ ✓

$\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ \frac{10}{3} \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ \frac{10}{3} \end{bmatrix}$ $\begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{5}{3} & \frac{5}{3} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ ✓

10) $\frac{10}{3}$ because it has the highest variance. ✓

11) A centered $\mu = \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ✓

12) $\begin{bmatrix} \frac{-1.5-1.5}{\sqrt{2}} = -\frac{3\sqrt{2}}{2} \\ \frac{-0.5-0.5}{\sqrt{2}} = -\frac{\sqrt{2}}{2} \\ \frac{0.5+0.5}{\sqrt{2}} = \frac{\sqrt{2}}{2} \\ \frac{1.5+1.5}{\sqrt{2}} = \frac{3\sqrt{2}}{2} \end{bmatrix}$ ✓